American Computer Science League

2023-2024 ● Contest 1: Short Problems ● Elementary Division

1. Computer Number Systems

Convert 1011010_2 to base 10.

A. 250
B. 132
C. 116
D. 90
E. 45

2. Computer Number Systems

Which of the following has the least number of 0s in its binary representation (not including any leading 0s)?

BED_{16} CAF_{16} 7356_8 2741_8

A. BED
B. CAF
C. 7356
D. 2741
E. more than 1 of them

3. Computer Number Systems

What is the number 980_{10} in base 8?

A. 640
B. 1724
C. 4271
D. 1632
E. 1742

4. Computer Number Systems

Given that you are living in a hexadecimal world and you have $4D5C_{16}$ people voting for one candidate and $3FA7_{16}$ voting for the other candidate in the election, how many total people voted for one of these two candidates in the election?

A. 7D03
B. 7CF3
C. 8DF3
D. 8D03
E. 8C04

5. Computer Number Systems

Colors can be expressed in RGB as either hex characters or decimal values between 0 and 255 inclusive. What are the 3 decimal equivalents to the hexadecimal color code "#241571" for Berry?

A. (36,19,114)
B. (37,20,112)
C. (36,21,113)
D. (35,21,113)
E. (36,21,112)

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1. 计算机数字系统

将 1011010_2 转换为 10 进制。

A. 250
B. 132
C. 116
D. 90
E. 45

2. 计算机数字系统

下列哪一项的二进制表达中 0 的个数最少 (不包括任意前导 0)?

BED_{16} CAF_{16} 7356_8 2741_8

A. BED
B. CAF
C. 7356
D. 2741
E. 以上选项中有不止一个正确答案

3. 计算机数字系统

980_{10} 用八进制表示是多少？

A. 640
B. 1724
C. 4271
D. 1632
E. 1742

4. 计算机数字系统

假设在一个十六进制的世界中，有 $4D5C_{16}$人给一个选举候选人投票，有 $3FA7_{16}$ 人给另一个选举候选人投票，请问一共有多少人为这两位候选人进行了投票？

A. 7D03
B. 7CF3
C. 8DF3
D. 8D03
E. 8C04

5. 计算机数字系统

RGB 颜色既可以用十六进制表示，也可以用位于 0 - 255 之间的（包括 0 和 255）十进制表示。与 Berry 的十六进制代码 “#241571” 等值的 3 个十进制数是多少？

A. (36,19,114)
B. (37,20,112)
C. (36,21,113)
D. (35,21,113)
E. (36,21,112)